
Highlights

An Explainable Rule-Based Approach to Automatic Rhythm Game Chart Generation

- Explainable rule-based approach generates rhythm-game charts from any audio.
- Lightweight NumPy MIR pipeline runs on CPU; a 3.5-min song in ~1.4 seconds.
- Feature-based pattern scoring with contour motif memory and density control.
- Reproducible under a fixed seed with controlled probabilistic variation.
- No GPU, no training data, and no copying of any charter's work; MIT-licensed.